Path-Connectedness of Scale-Invariant Floer-Hessian Growth Strata

Candidate partial result for arXiv:2408.00269

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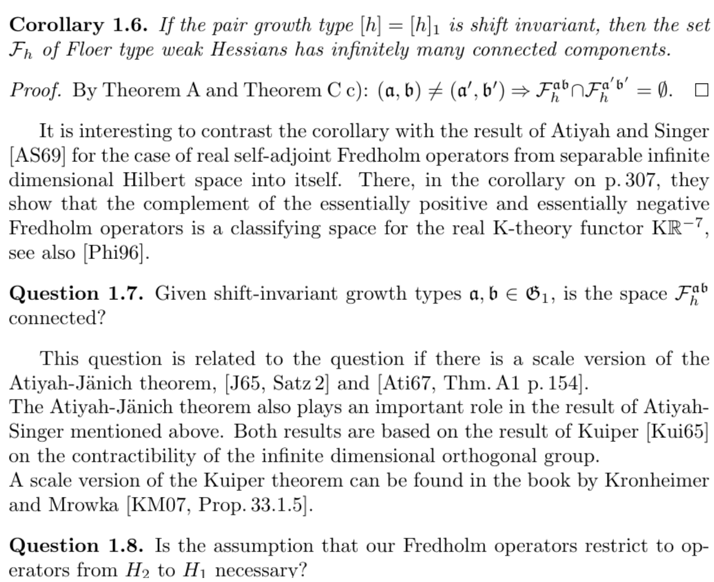
Status: candidate partial result, likely valid. Question 1.7 remains open for general shift-invariant, non-scale-invariant pair growth.

1 Source question and notation

Frauenfelder and Weber ask the following in [1, Question 1.7], on arXiv PDF page 6.

> **Question 1.7.** Given shift-invariant growth types $\mathfrak{a}, \mathfrak{b} \in \mathfrak{G}_1$, is the space $\mathcal{F}_h^{\mathfrak{ab}}$ connected?



Corollary 1.6. *If the pair growth type $[h] = [h]_1$ is shift invariant, then the set $\mathcal{F}_h$ of Floer type weak Hessians has infinitely many connected components.*

Proof. By Theorem A and Theorem C c): $(\mathfrak{a}, \mathfrak{b}) \neq (\mathfrak{a}', \mathfrak{b}') \Rightarrow \mathcal{F}_h^{\mathfrak{ab}} \cap \mathcal{F}_h^{\mathfrak{a}'\mathfrak{b}'} = \emptyset$. □

It is interesting to contrast the corollary with the result of Atiyah and Singer [AS69] for the case of real self-adjoint Fredholm operators from separable infinite dimensional Hilbert space into itself. There, in the corollary on p. 307, they show that the complement of the essentially positive and essentially negative Fredholm operators is a classifying space for the real K-theory functor $\mathrm{K}\mathbb{R}^{-7}$, see also [Phi96].

Question 1.7. Given shift-invariant growth types $\mathfrak{a}, \mathfrak{b} \in \mathfrak{G}_1$, is the space $\mathcal{F}_h^{\mathfrak{ab}}$ connected?

This question is related to the question if there is a scale version of the Atiyah-Jänich theorem, [J65, Satz 2] and [Ati67, Thm. A1 p. 154].
The Atiyah-Jänich theorem also plays an important role in the result of Atiyah-Singer mentioned above. Both results are based on the result of Kuiper [Kui65] on the contractibility of the infinite dimensional orthogonal group.
A scale version of the Kuiper theorem can be found in the book by Kronheimer and Mrowka [KM07, Prop. 33.1.5].

Question 1.8. Is the assumption that our Fredholm operators restrict to operators from H_2 to H_1 necessary?

Figure 1: Question 1.7 and its context in arXiv:2408.00269, PDF page 6.

We use the definitions of [1]. Thus

$$H_r = \ell^2_{h^r}, \qquad r \in \mathbb{R},$$

and a weak Hessian is an H_0-symmetric Fredholm operator of index zero $A : H_1 \to H_0$ whose restriction $A : H_2 \to H_1$ is also Fredholm of index zero. The subspace $\mathcal{F}_h^{\mathfrak{ab}}$ consists of the Floer-type weak Hessians whose negative and positive eigenvalue growth types are $\mathfrak{a}$ and $\mathfrak{b}$, in the translation-invariant sense of [1, Section 1.1].

A growth type $[h]$ is *scale-invariant* when it has a representative satisfying

$$h(2n) \le Ch(n) \qquad (n \in \mathbb{N}) \tag{1}$$

for some C. This is stronger than shift invariance; polynomial growth is scale-invariant, whereas exponential growth is shift-invariant but not scale-invariant [1, Section 2.2.2].

2 Proof intuition

Translate two Hessians slightly so that both become invertible. Their positive eigenvalues have the same growth type, as do their negative eigenvalues. Match positive eigenvectors to positive eigenvectors by rank, and do the same on the negative side. The resulting orthogonal operator is not merely an H_0-unitary: the graph norms of the two Hessians show that it and its inverse are bounded on H_1 and H_2.

The key use of scale invariance is topological. Applied to the Hilbert pair (H_0, H_2), whose weight is h^2, the mild-spectrum Kuiper theorem connects the matching operator to the identity through orthogonal maps bounded on H_2. Conjugating along this path aligns the two eigenbases while staying inside the weak-Hessian topology. In the common eigenbasis, linearly interpolate corresponding eigenvalues. Since matched eigenvalues have the same sign and comparable magnitudes, no eigenvalue reaches zero and both Fredholm-level estimates remain uniform.

3 Two lemmas

Lemma 1 (Scale-orthogonal path)**.** *Assume* (1)*. If U is orthogonal on H_0 and U, U^{-1} restrict to bounded operators on H_2, then there is a path $(U_t)_{0\le t\le 1}$ from the identity to U such that every U_t is orthogonal on H_0, U_t, U_t^{-1} are bounded on H_2, and the path is continuous in the pair operator topology. Consequently it is continuous on H_1 as well.*

Proof. The Hilbert pair (H_0, H_2) is isometric to $(\ell^2, \ell^2_{h^2})$. Its pair weight is mild because

$$h(2n)^2 \le C^2 h(n)^2.$$

The mild-spectrum Kuiper theorem [2, Proposition 33.1.5] says that the group of pair-unitaries is contractible. Its proof is Kuiper's argument in the pair-bounded operator algebra; the same proof over the real field replaces unitary maps by orthogonal maps, as in the real form of Kuiper's theorem [3]. This supplies (U_t). Boundedness and norm-continuity on H_1, the midpoint space of the Hilbert scale, follow by interpolation between H_0 and H_2. $\square$

Lemma 2 (Sign-preserving eigenbasis matching)**.** *Let $A, B \in \mathcal{F}_h^{\mathfrak{ab}}$ be invertible. There is an H_0-orthogonal operator U which maps the positive eigenvectors of A to those of B in increasing spectral order and likewise on the negative side, and for which U, U^{-1} are bounded on H_1 and H_2.*

Proof. Choose H_0-orthonormal eigenbases

$$Av_n^+ = \alpha_n^+ v_n^+, quad Av_n^- = -\alpha_n^- v_n^-, \qquad Bw_n^+ = \beta_n^+ w_n^+, quad Bw_n^- = -\beta_n^- w_n^-,$$

where all displayed eigenvalue magnitudes are positive and are nondecreasing, with multiplicities. Define $Uv_n^\pm = w_n^\pm$. This is orthogonal on H_0.

Equality of the signed growth types gives a constant $K \geq 1$ such that

$$K^{-1}\beta_n^{\pm} \leq \alpha_n^{\pm} \leq K\beta_n^{\pm} \qquad (n \in \mathbb{N}). \tag{2}$$

If the source convention records squared eigenvalues, take square roots of its comparison constants to obtain (2).

Because $A : H_1 \to H_0$ is an isomorphism, its graph norm is equivalent to the fixed H_1 norm. Hence, for finite eigenvector sums,

$$\|x\|_1^2 \asymp \|Ax\|_0^2 = \sum_{n,\varepsilon \in \{+,-\}} (\alpha_n^{\varepsilon})^2 |x_n^{\varepsilon}|^2. \tag{3}$$

The analogous formula holds for B in the w-basis. Equations (2)–(3) show that U and U^{-1} are bounded on H_1.

The regularity theorem of [1] says that invertibility on $H_1 \to H_0$ is equivalent to invertibility of the restriction $A : H_2 \to H_1$. Applying the level-one graph-norm equivalence twice gives

$$\|x\|_2 \asymp \|Ax\|_1 \asymp \|A^2 x\|_0, \tag{4}$$

and therefore, again first on finite sums,

$$\|x\|_2^2 \asymp \sum_{n,\varepsilon \in \{+,-\}} (\alpha_n^{\varepsilon})^4 |x_n^{\varepsilon}|^2.$$

The corresponding formula for B, together with (2), proves boundedness of U and U^{-1} on H_2. Density extends the estimates to the full spaces. $\square$

4 Partial answer to Question 1.7

Theorem 3. *Let the pair growth type $[h]$ be scale-invariant, and let $\mathfrak{a}, \mathfrak{b}$ be shift-invariant growth types. Every nonempty space $\mathcal{F}_h^{\mathfrak{ab}}$ is path-connected in the topology induced by*

$$\mathcal{L}(H_1, H_0) \cap \mathcal{L}(H_2, H_1).$$

Proof. Take $A, B \in \mathcal{F}_h^{\mathfrak{ab}}$. The spectrum of each weak Hessian is discrete and has nonempty resolvent. Choose $\lambda_A \notin \operatorname{spec} A$ and $\lambda_B \notin \operatorname{spec} B$. The translation paths

$$s \longmapsto A - s\lambda_A \iota, \qquad s \longmapsto B - s\lambda_B \iota$$

remain in $\mathcal{F}_h^{\mathfrak{ab}}$ by the translation invariance proved in [1, Section 1.1], and their endpoints $A' = A - \lambda_A \iota$ and $B' = B - \lambda_B \iota$ are invertible. It is therefore enough to connect A' and B'.

Apply Lemma 2 and let U be the sign-preserving eigenbasis matching operator. Lemma 1 supplies a path U_t from the identity to U. Put

$$A_t = U_t A' U_t^{-1}.$$

Every A_t is H_0-symmetric because U_t is H_0-orthogonal. Since U_t and its inverse preserve H_1 and H_2, conjugation preserves Fredholm invertibility on both $H_1 \to H_0$ and $H_2 \to H_1$. The signed spectrum is unchanged, so $A_t \in \mathcal{F}_h^{\mathfrak{ab}}$. The operator path is continuous in both defining norms. Thus A' is connected to $\widetilde{A} := UA'U^{-1}$.

By construction, $\widetilde{A}$ and B' are diagonal in the same H_0-orthonormal eigenbasis, with corresponding eigenvalues of the same sign. Write these signed eigenvalues as a_j and b_j. Their magnitudes obey

$$K^{-1}|b_j| \leq |a_j| \leq K|b_j|.$$

For $0 \leq s \leq 1$, define

$$C_s = (1-s)\widetilde{A} + sB'.$$

On the common eigenbasis the eigenvalues are $c_j(s) = (1-s)a_j + sb_j$. Same signs and the comparison above imply

$$\min\{1, K^{-1}\}|b_j| \leq |c_j(s)| \leq \max\{1, K\}|b_j|.$$

Consequently C_s is an isomorphism $H_1 \to H_0$ and, using the squared graph-norm comparison, an isomorphism $H_2 \to H_1$. It is H_0-symmetric, of Floer type, and its positive and negative eigenvalue growth types remain $\mathfrak{a}$ and $\mathfrak{b}$. Hence $C_s \in \mathcal{F}_h^{\mathfrak{ab}}$ throughout. This path joins $\widetilde{A}$ to B'.

Concatenate these three paths with the reverse translation from B' to B. The resulting path lies entirely in $\mathcal{F}_h^{\mathfrak{ab}}$. □

5 Limitations and novelty check

The theorem does not prove connectivity for all shift-invariant growth. The scale-Kuiper input requires mild spectrum. For (H_0, H_2) that hypothesis is precisely supplied by scale invariance of h; shift invariance alone does not imply it. In particular the exponential example $h(n) = e^n$ from [1] lies outside the proof.

A bounded web and arXiv search on 11 August 2026 used the exact paper title, the wording of Question 1.7, and the phrases "Floer Hessian connected", "scale invariant Floer Hessian", and "fixed signed growth stratum". It located the source paper and later citations concerned with different Fredholm questions, but no paper claiming either a solution to Question 1.7 or the scale-invariant theorem above. Novelty confidence is therefore bounded rather than definitive.

6 Verification and human-review focus

The proof was checked at four sensitive points: mildness of the pair (H_0, H_2); the H_2 graph-norm identity; real-field applicability of the scale-Kuiper argument; and preservation of both weak-Hessian levels during eigenvalue interpolation. No computational experiment is used.

Human review should focus first on Lemma 1, specifically the real orthogonal form of the cited pair-Kuiper theorem, and then on (4). Once those are accepted, the path construction is explicit.

References

[1] U. Frauenfelder and J. Weber, *Growth of eigenvalues of Floer Hessians*, arXiv:2408.00269, 2024.

[2] P. B. Kronheimer and T. S. Mrowka, *Monopoles and Three-Manifolds*, Cambridge University Press, 2007; Proposition 33.1.5.

[3] N. H. Kuiper, *The homotopy type of the unitary group of Hilbert space*, Topology 3 (1965), 19–30.